

Motivations

- Modern datacenters are becoming increasingly diverse and complex
- Datacenters are made of a wide range of components spanning IT, cooling, and power distribution infrastructures
- Traditional study of datacenter designs has focused on the IT equipment, which is no longer adequate
- There is a notable lack of tools capable of generating realistic datacenter hardware configurations

Objectives

- Develop a generative tool for IT configurations that creates realistic designs for four datacenter types
- Develop a generative tool for datacenter "Gray Space" configurations, including power and cooling designs to match IT requirements.
- Project future datacenter configurations
- Provide metrics to support "what-if" design analysis
- Enable research on datacenters

Methodology

- Define datacenter use-cases: AI training, AI inference, Cloud
- Collect reference IT hardware configurations (Microsoft, Meta, NVIDIA, IBM, ...), and systems (xAI Colossus, El Capitan, DGX SuperPOD, ...)
- Classify nodes and datacenters

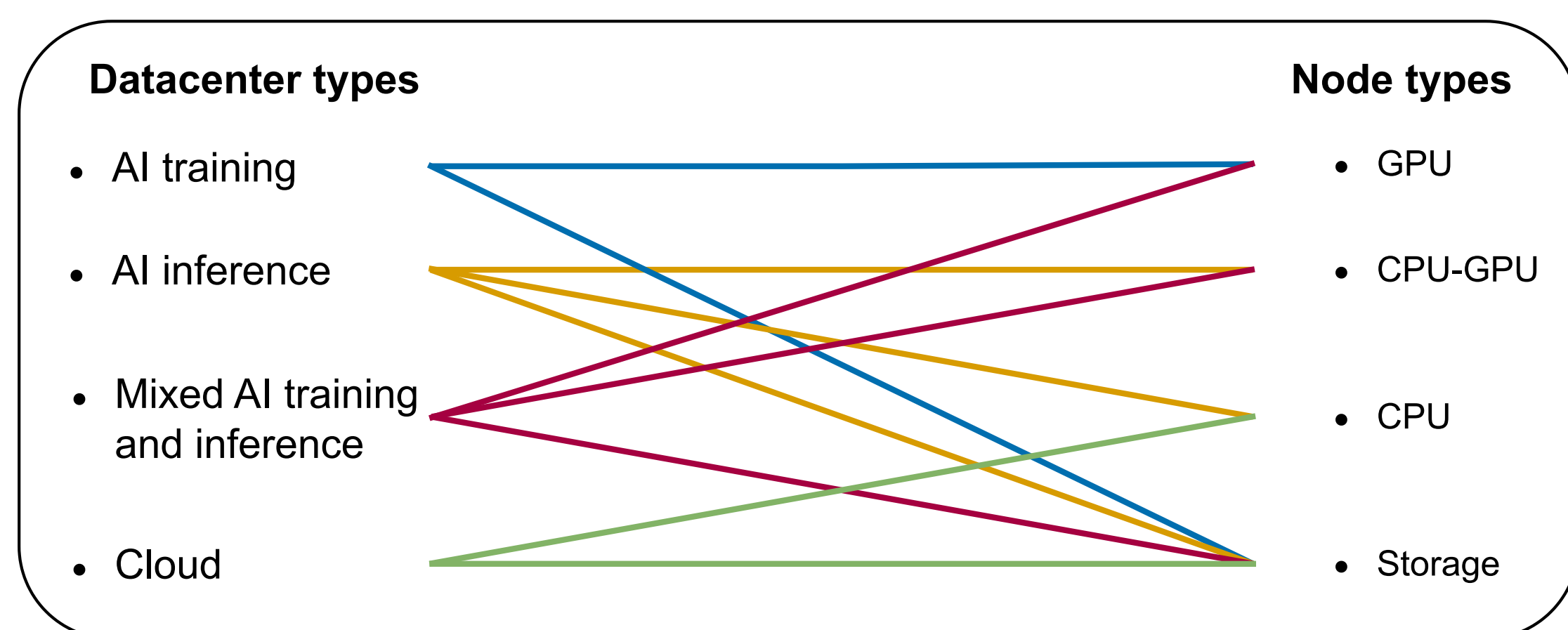


Figure 1. Datacenter use cases

- Aggregate types into a canonical model for IT racks
- Collect reference cooling and power distribution Components
 - Cooling: CDUs, water-cooled chillers, dry coolers, Evaporative cooling towers
 - Power distribution: PDUs, UPSs, MSBs, Backup generators
- Construction of Generative model – Configures with objective
 - Minimize space
 - Minimize power

DCGen: generative tool

- Generates representative datacenter hardware configurations
 - IT hardware
 - Cooling hardware
 - Power distribution systems
- Provides critical metrics
 - IT power density
 - IT and non-IT (cooling, power distribution systems) power demand
 - Space requirements – White space (IT) and Gray space (cooling and power distribution systems)

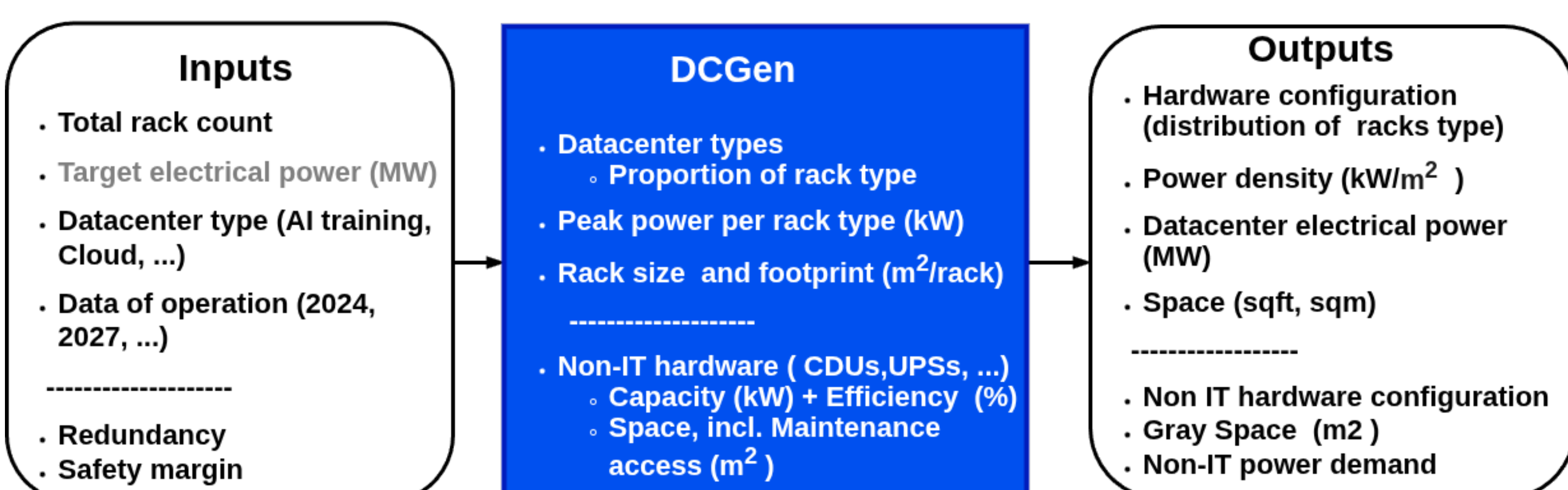


Figure 2. DCGen takes either a desired number of racks or a target electrical power as input. It then produces hardware configurations and other crucial metrics

Example of Single Datacenter configuration based on xAI COLOSSUS

- Goal:** show how DCGen generate datacenter IT hardware configuration
- Scenarios:**
 - Original xAI Colossus design (1,563 × 100kW GPU and 203 × 34kW storage racks)
 - Datacenter with twice xAI Colossus' compute capacity (rack count)

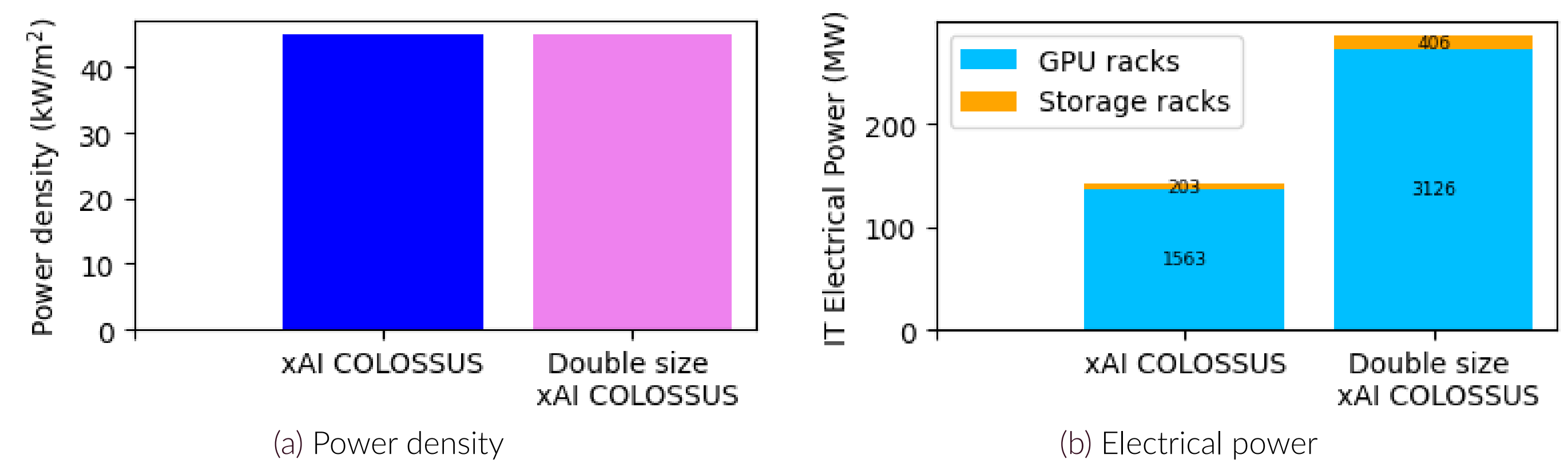


Figure 3. Example of single datacenter configurations.

- Observations**
 - Power density is not impacted by rack count
 - Racks distribution and Space utilization scale up proportionally

Case study: Power density and load

- Scenario:** Fixed number of racks (10,000)

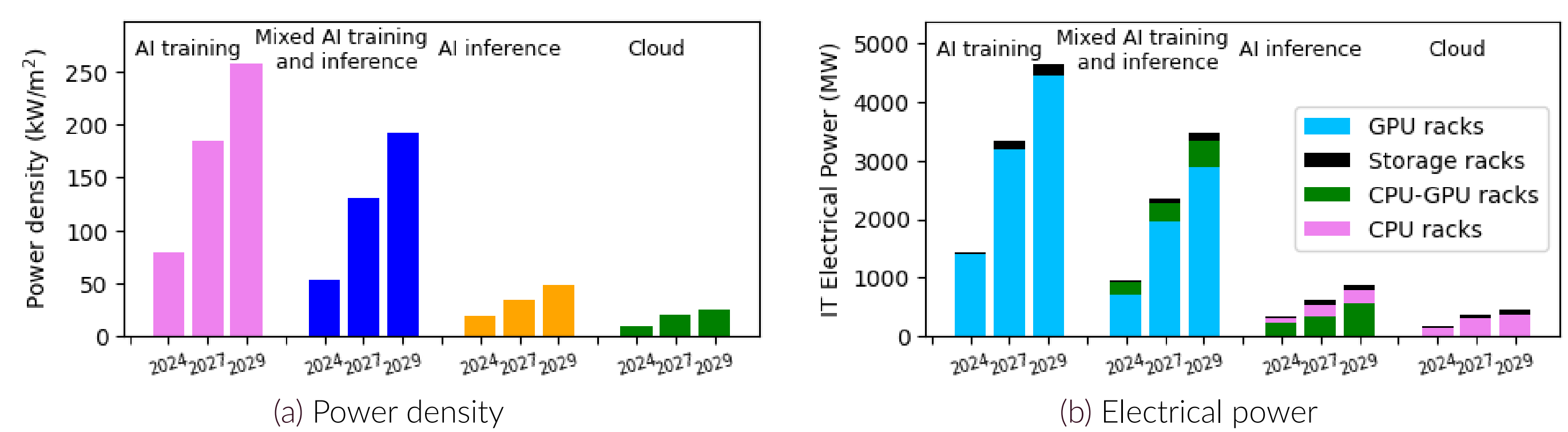


Figure 4. 10,000-rack datacenters configurations.

- Observations**

Type	Today's DC	2027 DC	2029 DC
AI training datacenter	Power density (kW/m ²) 79.8 IT Power (MW) 1,4000	2.3× Today	3.2× Today
Mixed AI training and inference datacenter	53.7 (1.5× lower)	963.3	2.4× Today
AI inference datacenter	18.8 (4.2× lower)	338.5	1.8× Today
Cloud datacenter	10.4 (7.7× lower)	186.5	2× Today

Table 1. 10,000-racks datacenter configurations

Case Study: Space requirements

- Scenario:** Fixed Electrical power (1GW) today, same IT racks in the future

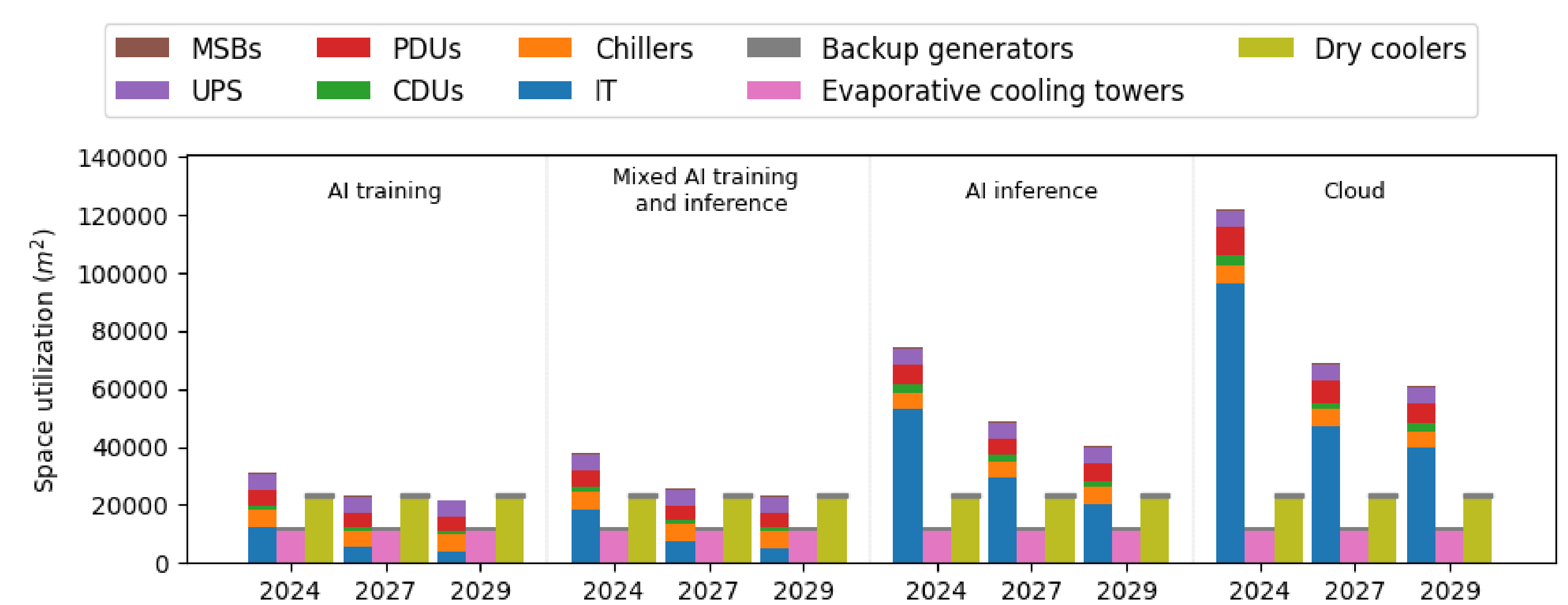


Figure 5. IT and Gray Space in 1GW datacenters: indoor (first bars), outdoor space in evap. cooling (second bars), outdoor space in dry cooling (third bars)

- Observations**
 - Gray space inside datacenter = IT space today (AI training, Mixed AI training and Inference)
 - Space efficient design: Outdoor space not a major concern today
 - 2027 and 2029 AI training: cooling and power distribution power dominates inside and outside when dry cooling is employed

Acknowledgments

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